Auditable Neural Martingale Control Variates for Basket Option Pricing

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Abstract

Small Monte Carlo standard errors do not establish accurate option prices when a learned control variate is centered at an incorrect expectation. We reconstruct a neural PDE pricing project whose saved one-dimensional hybrid estimate is 8.95% below its analytical reference despite a reported standard error of 4.32×10^{-7}. We identify the centering failure, separate inherited artifacts from new measurements, and implement predictable neural martingale controls with independent pilot fitting and explicit reference uncertainty. The resulting study compares smooth Deep Galerkin and multilayer-perceptron surrogates with plain Monte Carlo, antithetic sampling, geometric-basket controls, and scrambled Sobol integration. Analytical calibration isolates the effects of control resolution and inaccurate gradients from approximation bias. Our contribution is an auditable implementation and evaluation protocol, not a new martingale identity or an assertion of state-of-the-art performance. The finite-budget results include 18 freshly trained models and 1,872 pricing estimates. At the central scenarios, DGM controls reduce estimated standard errors by approximately 4.5–5.4 times relative to plain Monte Carlo, but cheap geometric controls are substantially faster. The study therefore supports estimator validity, not a claim of practical neural superiority.

1 Introduction

Neural PDE solvers offer mesh-free function approximations in settings where tensor-product spatial grids are impractical. A learned price, however, remains an approximation. Automatic differentiation returns derivatives of that approximation, not necessarily accurate financial sensitivities. Monte Carlo control variates offer a complementary use of a learned model: preserve the target expectation while altering sampling variance. This separation is useful only when the control's expectation is genuinely known.

We investigate this distinction in an inherited basket-pricing project. Its near-terminal network predictions were highly correlated with realized payoffs, but were centered at an unverified time-zero network prediction. A small residual sample variance then coexisted with a substantial price error. Other inherited tables mixed relative L^2 and mean relative errors, and some aggregate entries were rescaled from different sampling budgets. These findings motivate a reconstruction rather than cosmetic revision of the original claims.

Our contributions are threefold: (i) an explicit diagnostic counterexample and source-level audit; (ii) a self-contained smooth PDE implementation and independently piloted martingale estimator; and (iii) an immutable, multi-seed screening study with strong classical comparators, analytical calibration, and cost accounting. We do not equate unbiasedness, variance reduction, computational efficiency, novelty, and industrial readiness: each requires separate evidence.

2 Related work and scope

The Deep Galerkin Method (DGM) of Sirignano and Spiliopoulos [4] trains a neural approximation using differential-equation residuals and condition losses. We use its gated architecture as a baseline, not as a new architectural contribution. Deep BSDE methods [1] offer a probabilistic alternative; we discuss but do not claim to reproduce them in the new experiments.

Most directly, Sabate Vidales, Šiška and Szpruch [3] combine approximate neural gradients with martingale controls, including high-dimensional experiments. Hinds and Tretyakov [2] study neural variance

reduction for Brownian and Lévy-driven SDEs, including direct variance-based learning. Consequently, reframing a neural PDE solution as a control variate is not itself new. Our narrower contribution is a reproducible analysis of failure modes and the incremental value of neural controls after a cheap analytic control. Neither the classical projection formulas below nor conditional unbiasedness is asserted as a new theorem.

3 Problem and failure diagnosis

Let $x_i = \log(S_i/K)$ and

$$dx_t = \mu\,dt + B\,dW_t, \qquad \mu_i = r - \tfrac{1}{2}\sigma_i^2, \quad BB^\top = A, \quad A_{ij} = \rho_{ij}\sigma_i\sigma_j. \tag{1}$$

The arithmetic basket payoff is $\phi(x) = K(d^{-1}\sum_i e^{x_i} - 1)^+$ and the target price is $v = \mathbb{E}[Y]$, where $Y = e^{-rT}\phi(x_T)$. The log-coordinate pricing operator is

$$\mathcal{L}u = \partial_t u + \mu^\top \nabla_x u + \tfrac{1}{2}\operatorname{tr}(A\nabla_x^2 u) - ru. \tag{2}$$

3.1 Why terminal correlation is insufficient

For a control C centered at a proposed mean m and a fixed coefficient b,

$$\mathbb{E}[Y - b(C - m)] - v = -b(\mathbb{E}[C] - m). \tag{3}$$

High correlation says nothing about the right-hand side. In particular, take $C = \phi(x_T)$, $m = v$, and $b = e^{-rT}$. Then every adjusted observation is exactly $e^{-rT}v$: the sample variance vanishes, but the estimator is biased unless $rT = 0$ or $v = 0$. Even using the exact terminal payoff does not repair the incorrect mean.

For a sufficiently regular approximate solution u, Itô's formula gives

$$\mathbb{E}[e^{-rT}u(T, x_T)] = u(0, x_0) + \mathbb{E}\int_0^T e^{-rt}\mathcal{L}u(t, x_t)\,dt. \tag{4}$$

Thus even after correcting discounting, the initial value of an approximate PDE solution is not generally the expectation of its discounted terminal value.

Inherited evidence, not a new experiment. The saved 1D hybrid price is 0.095156352459. The exact Black–Scholes value at $S_0 = K = 1$, $r = 0.05$, $\sigma = 0.2$, $T = 1$ is 0.104505835722. Their difference is approximately 21,638 reported hybrid standard errors. This calculation is independently reproduced by the audit command. No legacy performance value is imported into the new experiment tables.

3.2 Other corrected sources of invalid inference

The reconstruction removes a payoff lifting that switched between smooth and kinked functions at train/evaluation time; a zero-value boundary penalty that was false on multidimensional basket faces; and a stochastic trace estimate whose square optimized $R^2 + \mathrm{Var}(\widehat{R})$ rather than R^2. It also replaces individual-path antithetic uncertainty by pair-level uncertainty and eliminates test-based checkpoint selection. The full audit provides original file paths and distinguishes source inspection from reruns.

4 Corrected estimator and implementation

4.1 Predictable neural controls

Fix a model u_θ trained independently of evaluation paths. For a grid $0 = t_0 < \cdots < t_M = T$, define

$$C_\theta = \sum_{k=0}^{M-1} e^{-rt_k}\nabla_x u_\theta(t_k, x_{t_k})^\top B\Delta W_k. \tag{5}$$

The gradient is evaluated before the new Brownian increment. We use exact Gaussian log-price transitions; control discretization therefore changes variance but does not introduce payoff discretization bias in this model.

Proposition 1 (Conditional unbiasedness)**.** *Let $\mathcal{H}$ contain training and independent pilot information. Suppose the frozen integrands in* (5) *are predictable and square integrable conditional on $\mathcal{H}$. Let G be another integrable control with $\mathbb{E}[G \mid \mathcal{H}] = 0$, and let $\widehat{b}$ be a finite, $\mathcal{H}$-measurable coefficient vector. For fresh independent paths,*

$$\widehat{v} = N^{-1} \sum_{n=1}^{N} \left(Y_n - \widehat{b}_1 G_n - \widehat{b}_2 C_{\theta,n} \right) \tag{6}$$

satisfies $\mathbb{E}[\widehat{v} \mid \mathcal{H}] = v$. Unconditional unbiasedness follows when the adjusted observations are integrable.

Proof. Each summand of C_θ has zero conditional expectation by independence and zero mean of the fresh Brownian increment. Apply the tower property, the known zero mean of G, and measurability of $\widehat{b}$. □

This result does not guarantee variance reduction. For the exact price function and continuous-time integrand h^*, the martingale representation yields $Y - v = \int_0^T (h_t^*)^\top dW_t$. For any admissible stepwise approximation h, Itô isometry gives

$$\operatorname{Var}\left(Y - \int_0^T h_t^\top dW_t \right) = \mathbb{E} \int_0^T \|h_t^* - h_t\|^2 dt. \tag{7}$$

This identity explains why price accuracy and useful control gradients are related but distinct, and why deliberately bad gradients can increase variance.

4.2 Geometric control and incremental neural value

For $\phi_g(x) = K(e^{\bar{x}} - 1)^+$, use $G = e^{-rT}\phi_g(x_T) - v_g$, where v_g is an exact lognormal expectation. Writing $m = \bar{x}_0 + \bar{\mu}T$, $q = Td^{-2}\mathbf{1}^\top A\mathbf{1}$,

$$v_g = Ke^{-rT}\left[e^{m+q/2}\Phi((m+q)/\sqrt{q}) - \Phi(m/\sqrt{q}) \right], \tag{8}$$

with the deterministic limit used when $q = 0$.

With population covariance Σ_C and $c = \operatorname{Cov}(C, Y)$, optimal linear coefficients are $b^* = \Sigma_C^\dagger c$. To interpret adding a neural control, residualize it against G: $\widetilde{C} = C_\theta - \operatorname{Cov}(C_\theta, G)G/\operatorname{Var}(G)$. The population variance reduction beyond optimal geometric control is

$$\frac{\operatorname{Cov}(Y, \widetilde{C})^2}{\operatorname{Var}(\widetilde{C})}, \tag{9}$$

when the denominator is positive. A neural control redundant with G adds no population benefit. Independent pilot OLS approximates this projection using centered, standardized columns and an SVD. Pilot means are not used as evaluation centers. Finite-pilot estimation can worsen variance, so monotonic improvement is not claimed. The one-dimensional arithmetic and geometric payoffs coincide; their roundoff-level residual is an analytic baseline, not a neural achievement.

4.3 Smooth training and numerical safeguards

We train raw tanh DGM and MLP outputs with

$$J(\theta) = \mathbb{E}[|\mathcal{L}u_\theta|^2] + 10\mathbb{E}[|u_\theta(T, x) - \phi(x)|^2]. \tag{10}$$

There is no hard payoff embedding or artificial boundary-face loss. Interior collocation uses uniform time and Gaussian log prices with standard deviation 0.3. Terminal samples use the same spatial law and the exact payoff target. All Hessian contractions are deterministic and exact under automatic differentiation. This is a finite-budget training baseline, not a claim of global PDE convergence. Nonfinite losses and estimator records fail explicitly.

5 Experimental protocol

The completed protocol uses CPU float64, fixed Adam learning rate 10^{-3}, 300 steps, batch size 64, width 32, and depth 2. Each architecture is trained with seeds 11, 29 and 47 at dimensions 1, 5 and 10: 18 fresh models in total. No legacy checkpoint is loaded and no test labels choose a checkpoint. Equal width is not equal parameter count or training cost.

We fix $K = T = 1$, $r = 0.05$, $\sigma_i = 0.2$, $\rho_{ij} = 0.3$ for $i \neq j$. Three scenario centers are $m_0 \in \{0.85, 1, 1.15\}$; for $d > 1$ spots are $S_{0,i} = m_0 \exp(\text{linspace}(-0.12, 0.12, d)_i)$. These heterogeneous vectors avoid the equal-asset diagonal, but are not a full-dimensional test distribution. Plain MC, antithetic MC, geometric control and RQMC additionally run at dimensions 25, 50 and 100. Those extra dimensions provide classical context only.

Each estimate uses 1024 evaluation paths; fitted controls use another 1024 independent pilot paths. Neural controls use eight time steps. RQMC uses eight independent scrambles of 128 Sobol points. There are eight repeated evaluations per scenario/model. Architecture comparisons share evaluation randomness where their scenario and training-seed labels coincide. Evaluation repetitions are not additional independent training seeds.

One-dimensional prices use an exact analytical reference. Higher-dimensional arithmetic baskets use a separate 16-scramble, 32768-point-per-scramble RQMC reference, with its uncertainty retained. RMSE against this reference includes its unknown error; it is not labelled exact pricing RMSE. Sample standard errors use independent paths, antithetic pair means, or independent scramble means as appropriate, with sample variance and Student-t interval multipliers.

For calibration, geometric payoffs have exact prices and analytical gradients. We test grids 4, 16 and 64 in dimensions 1, 5 and 10 with 40 replications per setting. Additional gradient scales $-1, 0, 0.5, 2$ at eight steps deliberately degrade the control. These are labelled oracle diagnostics, not learned results. Coverage uncertainty is reported using Wilson intervals.

Cost and provenance. Evaluation, pilot and training time are recorded separately. Equal path counts are not equal compute: neural estimates require repeated gradient evaluations. We distinguish warm evaluation-plus-pilot time from cold-start cost including training. Hardware-specific timings are descriptive, not deployment guarantees. An earlier 600-step protocol was interrupted by an execution timeout. Its partial artifacts are retained locally but excluded from every new result table. The replacement budget was fixed before its run; it is a screening study. Run and checkpoint hashes, actual path counts, seeds, software versions and source fingerprints accompany the measurements.

6 Results

6.1 Pricing accuracy and computational cost

The completed main run contains 18 models, 1,872 pricing-estimate records, 54 direct surrogate prices, and 840 oracle-calibration records. Its recorded wall time is 753 seconds; this is an execution receipt, not a cross-hardware performance claim. Table 1 reports central scenarios. All three moneyness scenarios, individual repetitions and training-seed dispersion are retained in machine-readable artifacts.

At $d = 5$, DGM's pilot-fitted neural control has RMS SE approximately 5.94×10^{-4}, compared with 3.18×10^{-3} for plain MC. The combined control reduces it further to 2.70×10^{-4}, versus 3.24×10^{-4} for geometric control alone. This modest incremental variance reduction costs approximately 378 ms per combined estimate, compared with 3.75 ms for geometric control. At $d = 10$, the corresponding combined and geometric RMS SEs are 2.31×10^{-4} and 2.59×10^{-4}, with warm costs approximately 326 ms and 4.49 ms. These measurements do not establish a neural cost advantage. Training costs, recorded separately, further increase cold-start cost.

The three-state direct-price relative L^2 errors in Table 2 are materially larger than the inherited headline might suggest. DGM is more accurate than this equal-width MLP in the short runs, but the architectures differ in parameter count and compute, and neither has been shown to be converged. The table is not a global price-surface norm.

6.2 Analytical calibration and inaccurate controls

Table 3 separates control-grid effects from neural training. At 64 steps, the analytical-gradient controls attain mean variance ratios of approximately 216, 237 and 243 for dimensions 1, 5 and 10, respectively.

d	Method	Arch.	RMSE	RMS SE	Warm ms
1	Antithetic	CLASSICAL	1.42e-03	3.18e-03	4.6
1	Geometric CV	CLASSICAL	1.20e-17	6.42e-18	4.8
1	MC	CLASSICAL	4.03e-03	4.61e-03	0.7
1	RQMC	CLASSICAL	3.97e-04	4.35e-04	30.8
1	Combined CV	DGM	1.55e-17	8.07e-18	383.9
1	Neural pilot	DGM	6.72e-04	9.48e-04	383.9
1	Combined CV	MLP	1.58e-17	8.39e-18	88.8
1	Neural pilot	MLP	7.70e-04	1.04e-03	88.8
5	Antithetic	CLASSICAL	2.32e-03	1.89e-03	2.4
5	Geometric CV	CLASSICAL	3.33e-04	3.24e-04	3.7
5	MC	CLASSICAL	2.83e-03	3.18e-03	1.4
5	RQMC	CLASSICAL	1.83e-04	1.77e-04	19.8
5	Combined CV	DGM	2.17e-04	2.70e-04	377.6
5	Neural pilot	DGM	5.11e-04	5.94e-04	377.6
5	Combined CV	MLP	2.79e-04	3.00e-04	84.9
5	Neural pilot	MLP	6.69e-04	7.17e-04	84.9
10	Antithetic	CLASSICAL	4.88e-04	1.70e-03	2.0
10	Geometric CV	CLASSICAL	1.19e-04	2.59e-04	4.5
10	MC	CLASSICAL	2.76e-03	2.91e-03	2.0
10	RQMC	CLASSICAL	1.67e-04	2.08e-04	37.5
10	Combined CV	DGM	2.50e-04	2.31e-04	325.5
10	Neural pilot	DGM	5.22e-04	6.47e-04	325.5
10	Combined CV	MLP	2.40e-04	2.34e-04	117.2
10	Neural pilot	MLP	7.41e-04	8.02e-04	117.2

Table 1: Central scenario results. RMSE is measured against the independent reference; RMS SE is the root mean square of reported sampling SEs. Warm time includes evaluation and pilot fitting, but excludes neural training. Classical rows aggregate eight repetitions; neural rows summarize three training seeds with eight evaluation repetitions each. These are descriptive, unequal-compute comparisons. The exact one-asset geometric control reaches roundoff precision.

d	Architecture	Mean rel. L^2 (%)	Seed SD (pp)
1	DGM	2.04	1.06
1	MLP	11.81	6.87
5	DGM	4.27	0.97
5	MLP	15.89	5.31
10	DGM	11.08	5.20
10	MLP	21.53	1.25

Table 2: Direct surrogate pricing over the three declared spot vectors. Mean and sample SD are across the three training seeds; pp denotes percentage points. No evaluation-path replication is counted as an extra training seed.

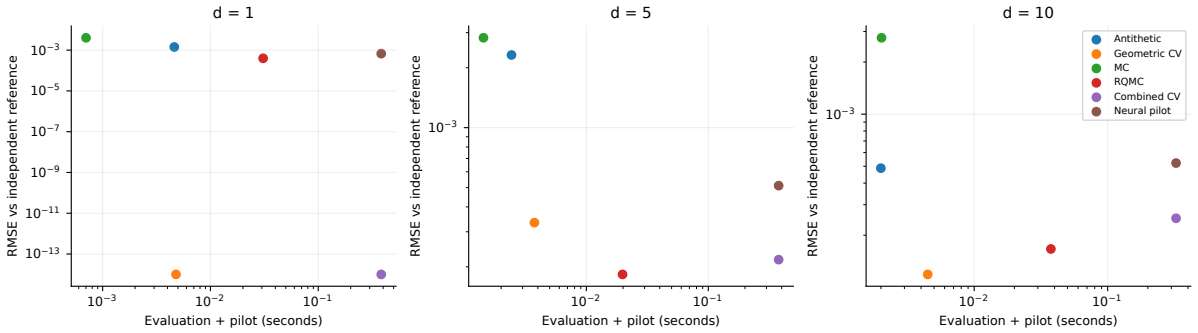


Figure 1: Central-scenario descriptive cost–accuracy comparison. Neural points use DGM; MLP results remain in the table and raw data. Warm cost includes pilot fitting. The one-dimensional roundoff-limited controls are plotted at a 10^{-14} visualization floor. This is not a matched-time budget sweep.

These are oracle results, not evidence that the learned network attains the same accuracy. Finer control grids reduce the residual variance without introducing bias into the exactly simulated payoff.

d	Steps	VR factor	Coverage	Wilson 95% interval
1	4	14.8	1.000	[0.912, 1.000]
1	16	56.0	0.975	[0.871, 0.996]
1	64	216.1	0.975	[0.871, 0.996]
5	4	16.6	0.925	[0.801, 0.974]
5	16	61.3	0.950	[0.835, 0.986]
5	64	236.9	0.925	[0.801, 0.974]
10	4	17.3	0.950	[0.835, 0.986]
10	16	62.4	0.950	[0.835, 0.986]
10	64	243.4	0.975	[0.871, 0.996]

Table 3: Analytical geometric-basket controls with coefficient one. VR factor is plain-payoff sample variance divided by adjusted sample variance, averaged over 40 replications. Coverage intervals describe uncertainty in the observed interval coverage, not in the price.

The deliberately sign-reversed gradient has variance ratio about 0.256, corresponding to roughly four times the plain-payoff variance. A zero gradient recovers the original Monte Carlo variance. This behavior contradicts any unconditional guarantee of variance reduction from an arbitrary network while remaining consistent with the zero-mean-control argument. The full scale sweep is retained, including negative results.

6.3 Independent learned-control and Greek diagnostics

An additional evaluation freezes the six one-dimensional checkpoints and runs 80 fresh repetitions per checkpoint, with 1024 pilot and 1024 evaluation paths. Both fixed and pilot-fitted neural-only controls are evaluated, producing 960 estimate records. No additional model is selected or trained using this test. Table 4 reports the pilot-fitted variant. Observed coverage ranges from 93.75% to 100%; one

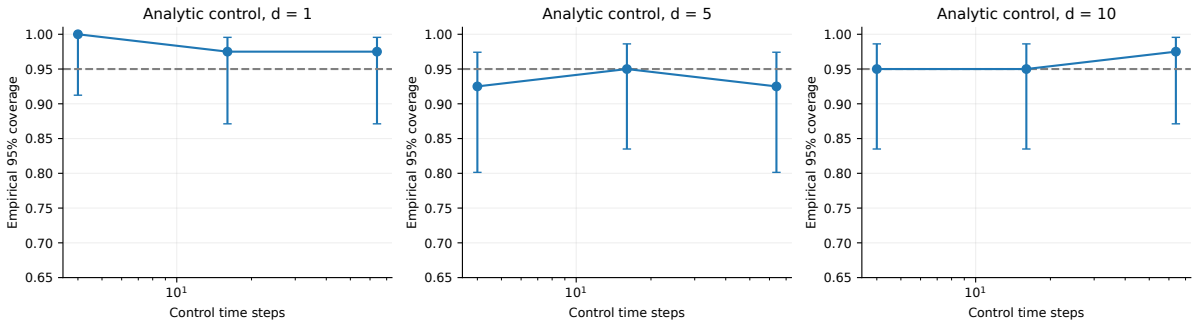


Figure 2: Oracle-control interval coverage with Wilson uncertainty bars and nominal 95% reference line. Forty repetitions per setting give wide intervals; the plot is a diagnostic, not a proof of finite-sample calibration.

configuration covers in all 80 replications, which does not establish its true coverage. Intervals remain broad. Each reported mean error is less than two corresponding Monte Carlo standard errors of that mean, which is a diagnostic consistency check, not a proof that finite-sample bias is zero.

Model / seed	Coverage	Wilson 95% interval	Bias	Bias MCSE
DGM / 11	0.9500	[0.878, 0.980]	-9.46e-05	1.16e-04
DGM / 29	0.9500	[0.878, 0.980]	1.63e-04	9.83e-05
DGM / 47	0.9625	[0.895, 0.987]	1.30e-04	9.49e-05
MLP / 11	0.9625	[0.895, 0.987]	7.95e-06	1.15e-04
MLP / 29	1.0000	[0.954, 1.000]	-1.42e-04	9.48e-05
MLP / 47	0.9375	[0.862, 0.973]	-1.25e-04	1.28e-04

Table 4: Learned neural-only control coverage against exact 1D prices, 80 independent repetitions per frozen checkpoint. Bias MCSE is uncertainty of the estimated bias across repetitions, distinct from each price estimate's SE.

We also evaluate price, Delta and Gamma on 101 spots from 0.6 to 1.4 at $t = 0, 0.5, 0.9$, for all six checkpoints: 1,818 pointwise records. Table 5 and Figure 3 show that smoothness alone does not imply accurate sensitivities. Near expiry, gamma errors are large. Negative prices and negative gamma values occur for some checkpoints and are recorded rather than hidden by plotting limits. These models should not be used as validated standalone pricing or risk engines.

Architecture	t	Price rel. L^2 (%)	Delta (%)	Gamma (%)
DGM	0.0	1.89	5.15	22.31
DGM	0.5	2.81	7.14	33.71
DGM	0.9	7.28	19.40	72.74
MLP	0.0	15.33	23.07	62.56
MLP	0.5	11.22	16.21	61.60
MLP	0.9	8.93	23.79	84.14

Table 5: Analytical 1D price and Greek validation over 101 spots, averaged across three training seeds. Each column reports its own relative L^2 error in percent.

7 Limitations and implications

Validity is not novelty or optimality. The estimator follows established martingale-control theory. Its conditional unbiasedness survives inaccurate frozen gradients, but neither optimal variance nor a practical speedup follows. A credible new method must improve on strong classical and neural variance-reduction baselines, not just plain MC.

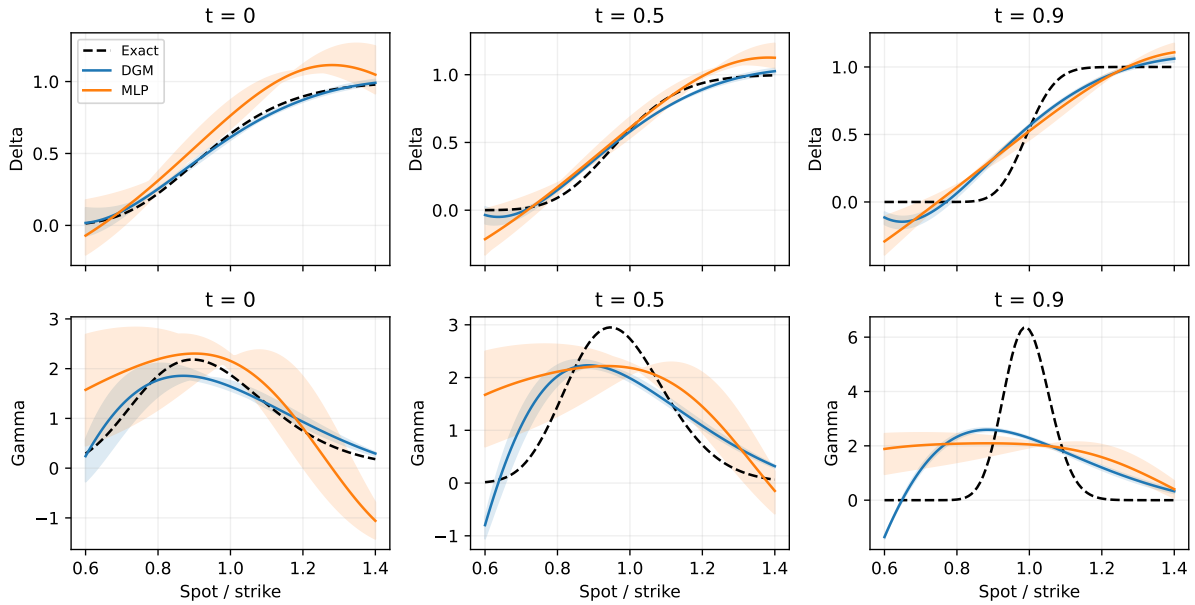


Figure 3: Analytical Greeks versus the mean learned Greeks; shading spans the three seed values, not a confidence interval. Negative values are not clipped. Near-expiry disagreement is retained as a substantive limitation.

Finite-budget scope. Three hundred training steps are insufficient to establish convergence. There are only three training seeds and three arithmetic-basket evaluation vectors per dimension. There is no neural scaling experiment beyond $d = 10$; the $d = 25, 50, 100$ results are classical-only. The main pricing model has fixed coefficients and a single payoff family. Market calibration, stochastic volatility, jumps, early exercise and out-of-distribution robustness remain unexamined. The analytical extension tests Greeks, but does not establish hedging performance.

Statistical and computational qualifications. Reference Monte Carlo error is retained, not eliminated. The coverage studies have finite replication uncertainty; the main-study RMSE comparisons are descriptive and do not support statistical dominance claims. Timing includes Python overhead and reflects one CPU machine with ordinary background activity, not a dedicated benchmark host. The Sobol construction is not tuned for effective dimension. Training amortization and matched-wall-time optimization have not been measured. Neither a finite-budget PDE approximation nor its automatic derivatives satisfy no-arbitrage requirements by construction.

A testable next contribution. Equation (9) suggests learning only the residual component unexplained by cheap analytic controls, then using independent pilot evidence to decide whether the extra neural cost is warranted. A convincing extension would need finite-sample selection analysis and matched-cost comparisons with direct integrand learning [3, 2]. We identify this as future work, not an accomplished SOTA result.

8 Conclusion

The reconstruction demonstrates why nearly zero sample variance can be misleading when a learned control is incorrectly centered. Predictable martingale controls repair the estimator's expectation without requiring a perfect PDE solution, and independent pilots avoid fitting coefficients on final evaluation paths. Fresh experiments confirm useful neural variance reduction but also reveal substantial computational overhead and imperfect direct prices and Greeks. The defensible outcome is a transparent research artifact and a sharper question about incremental neural value, not an industry-standard pricing method or a claim of conference-level novelty.

Artifact availability and research integrity. The isolated repository contains source, tests, configurations, completed-run manifests, checkpoint hashes, raw results and report generators. It retains a separate audit inventory of the original research files without modifying those files. Tables and figures are generated from hash-verified completed runs; the interrupted run and inherited aggregate performance are excluded. All data are synthetic. This draft retains the lead name from the inherited manuscript; final authorship, licensing and submission venue require owner confirmation.

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